

5. Find the weights required to perform the following classifications of given input patterns using the Hebb rule. The inputs are "1" where "+" is present "-1" where "." is present. The "L" pattern belongs to the target value +1 and the "U" pattern does not belong to the class (target value -1)

+	.	.									+	.	+
+	.	.									+	.	+
+	+	+									+	+	+

"L" "U"

Pattern	Inputs										Target y
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	b	
L	1	-1	-1	1	-1	-1	1	1	1	1	1
U	1	-1	1	1	-1	1	1	1	1	1	-1

Initially, $w_1 = w_2 = w_3 = w_4 = w_5 = w_6 = w_7 = w_8 = w_9 = b = 0$

Case 1

$$w_i(\text{new}) = w_i(\text{old}) + x_i y$$

$$w_1(\text{new}) = 0 + (1 \times 1) = 1$$

$$w_2(\text{new}) = 0 + (-1 \times 1) = -1$$

$$w_3(\text{new}) = 0 + (-1 \times 1) = -1$$

$$w_4(\text{new}) = 0 + (1 \times 1) = 1$$

$$w_5(\text{new}) = 0 + (-1 \times 1) = -1$$

$$w_6(\text{new}) = 0 + (-1 \times 1) = -1$$

$$w_7(\text{new}) = 0 + (1 \times 1) = 1$$

$$w_8(\text{new}) = 0 + (1 \times 1) = 1$$

$$w_9(\text{new}) = 0 + (1 \times 1) = 1$$

$$\Delta w_i = x_i y$$

$$\Delta w_1 = 1 \times 1 = 1$$

$$\Delta w_2 = -1 \times 1 = -1$$

$$\Delta w_3 = -1 \times 1 = -1$$

$$\Delta w_4 = 1 \times 1 = 1$$

$$\Delta w_5 = -1 \times 1 = -1$$

$$\Delta w_6 = -1 \times 1 = -1$$

$$\Delta w_7 = 1 \times 1 = 1$$

$$\Delta w_8 = 1 \times 1 = 1$$

$$\Delta w_9 = 1 \times 1 = 1$$

$$\Delta b = y = 1$$

$$b(\text{new}) = b(\text{old}) + y = 0 + 1 = 1$$

$$[\omega_1 \ \omega_2 \ \omega_3 \ \omega_4 \ \omega_5 \ \omega_6 \ \omega_7 \ \omega_8 \ \omega_9 \ b] = [1 \ -1 \ -1 \ 1 \ -1 \ -1 \ 1 \ 1 \ 1 \ 1]$$

Case II

$$\omega_1(\text{new}) = 1 + (1 \times -1) = 0$$

$$\omega_2(\text{new}) = -1 + (-1 \times -1) = 0$$

$$\omega_3(\text{new}) = -1 + (1 \times -1) = -2$$

$$\omega_4(\text{new}) = 1 + (1 \times -1) = 0$$

$$\omega_5(\text{new}) = -1 + (-1 \times -1) = 0$$

$$\omega_6(\text{new}) = -1 + (1 \times -1) = -2$$

$$\omega_7(\text{new}) = 1 + (1 \times -1) = 0$$

$$\omega_8(\text{new}) = 1 + (1 \times -1) = 0$$

$$\omega_9(\text{new}) = 1 + (1 \times -1) = 0$$

$$b(\text{new}) = 1 + -1 = 0$$

$$[\omega_1 \ \omega_2 \ \omega_3 \ \omega_4 \ \omega_5 \ \omega_6 \ \omega_7 \ \omega_8 \ \omega_9 \ b] = [0 \ 0 \ -2 \ 0 \ 0 \ -2 \ 0 \ 0 \ 0 \ 0]$$

$$\Delta \omega_1 = 1 \times -1 = -1$$

$$\Delta \omega_2 = -1 \times -1 = 1$$

$$\Delta \omega_3 = 1 \times -1 = -1$$

$$\Delta \omega_5 = -1 \times -1 = 1$$

$$\Delta \omega_6 = 1 \times -1 = -1$$

$$\Delta \omega_7 = 1 \times -1 = -1$$

$$\Delta \omega_8 = 1 \times -1 = -1$$

$$\Delta \omega_9 = 1 \times -1 = -1$$

$$\Delta b = y = -1$$

